

Fatima Jinnah Women University

“Project”

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ABSTRACT

Particular safety measures are suggested that a victim alone at home can practice self-applied first aid without provoking further injury. Mild wound injuries necessitate medical advice in order to determine the necessity of a tetanus shot. A physician should also inspect if the bandaging and medication was properly applied. If only one rescuer is available, there are general criteria which indicate that first aid be given in certain cases while other emergency situations necessitate urgent care. Various examples of emergency situations are illustrated which differentiate between life-threatening situation and less critical emergencies. An immediate evaluation of the severity of the victim's condition with corresponding type of treatment can save the victim's life. Drugs and medications should always be kept in a fixed place which is easily accessible to family members and eventually to medical personnel. We include a list of all necessary first aid materials, and their possible substitution in case they are not available. Fundamental regulations regarding gas leak-related accidents are listed. We advise not ringing the doorbell, electric lights should not be used, the windows and doors should be opened wide and the victim should be dragged away as far as possible. General indications are given in order to slow or decrease external bleeding.

Introduction

First aid is the initial and urgent care provided to anyone who has a small or significant disease or injury, with the goal of preserving life, preventing the condition from getting worse or fostering recovery. It involves both the full treatment of minor conditions, such as putting a plaster on a cut, as well as the initial intervention in a critical situation before professional medical care is available, such as administering cardiopulmonary resuscitation (CPR) while waiting for an ambulance. First aid is typically administered by a person with only little medical training. While psychological first aid is utilized as early therapy for persons who are at risk for developing post-traumatic stress disorder (PTSD), mental health first aid is an expansion of the first aid idea to address mental health. In Canada, a program me called Conflict First Aid is being tested. It aims to protect and restore a person's social or interpersonal well-being.

There are numerous instances that could necessitate first assistance, and many nations have laws, regulations, or other policies that specify a minimum standard of first aid supply in particular situations. This can include requiring first aid training in schools or requiring particular training or equipment (such an automated external defibrillator) to be provided in the workplace or at public gatherings. First aid, however, can include improvisation using materials on hand and is frequently performed by unskilled individuals without the need for special tools or experience.

1. First aid for heart attack

A heart attack is a medical emergency. Call 911 or the local emergency number if you think you or someone else is having a heart attack.

The average person waits 3 hours before seeking help for symptoms of a heart attack. Many heart attack patients die before they reach a hospital. The sooner the person gets to the emergency room, the better the chance of survival. Prompt medical treatment reduces the amount of heart damage.



1.1 Condition:

You feel a tight band of pain around your chest. The pain moves from your chest to your arms, shoulder, and neck. What could your pain mean? Could it be a heart attack, could it be the big one? Heart attacks are caused by interruption of blood supply to part of the heart. If the blood flow is blocked, your heart is starved of oxygen and heart cells die. A hard substance called plaque can build up in the walls of your coronary arteries. This plaque is made up of cholesterol and other cells. A heart attack can occur as a result of plaque buildup or the rupture of one of these plaques. We're not sure why heart attacks occur when they do. You may have a heart attack when you are resting or asleep, or after a sudden increase in physical activity,

when you are outside in cold weather, or after a sudden, severe emotional or physical stress, including an illness. So, how is a heart attack treated? If you go to the hospital for a suspected heart attack, a doctor or nurse will listen to your chest with a stethoscope. You will have a blood test to look for heart damage. A coronary angiography test can show your doctor how well blood is moving through your heart. If blood moves slowly, or not at all through your coronary arteries, you have either a narrowed, or blocked artery. Other tests can look at the valves and chambers of your heart and check for abnormal heart rhythms. If you've had a heart attack, doctors can do an emergency procedure called angioplasty. This surgery or procedure can open narrowed or blocked blood vessels. Usually they'll place a small, metal mesh tube, called a stent, in your artery to help keep it open. You may also receive drugs to break up the clot in your artery. Sometimes, doctors will do heart bypass surgery to get blood flowing to your heart muscle again. After you are treated in the hospital for a heart attack, you may need to take medicines to thin your blood, to protect your heart, or to improve your cholesterol levels. You may need to take these medicines for the rest of your life.

1.2 Causes:

A heart attack occurs when the blood flow that carries oxygen to the heart is blocked. The heart muscle becomes starved for oxygen and begins to die.

1.3 Symptoms:

Symptoms of a heart attack can vary from person to person. They may be mild or severe. Women, older adults, and people with diabetes are more likely to have subtle or unusual symptoms.

Symptoms in adults may include:

- Changes in mental status, especially in older adults.
- Chest pain that feels like pressure, squeezing, or fullness. The pain is most often in the center of the chest. It may also be felt in the jaw, shoulder, arms, back, and stomach. It can last for more than a few minutes, or come and go.
- Cold sweat.
- Lightheadedness.
- Nausea (more common in women).
- Indigestion.
- Vomiting.
- Numbness, aching, or tingling in the arm (usually the left arm, but the right arm may be affected alone, or along with the left).
- Shortness of breath.
- Weakness or fatigue, especially in older adults and in women.

1.4 First Aid:

If you think someone is having a heart attack:

- Have the person sit down, rest, and try to keep calm.

- Loosen any tight clothing.
- Ask if the person takes any chest pain medicine, such as nitroglycerin for a known heart condition, and help them take it.
- If the pain does not go away promptly with rest or within 3 minutes of taking nitroglycerin, call for emergency medical help.
- If the person is unconscious and unresponsive, call 911 or the local emergency number, then begin CPR.
- If an infant or child is unconscious and unresponsive, perform 1 minute of CPR, then call 911 or the local emergency number.

Many experts recommend chewing and swallowing a full dose aspirin, after calling 911 or the local emergency number. Only do this if you are not allergic to aspirin and do not have any condition that could make taking aspirin dangerous for you. The 911 operator can help you decide whether or not to take aspirin.

1.5 Do Not:

- DO NOT leave the person alone except to call for help, if necessary.
- DO NOT allow the person to deny the symptoms and convince you not to call for emergency help.
- DO NOT wait to see if the symptoms go away.
- DO NOT give the person anything by mouth unless a heart medicine (such as nitroglycerin) has been prescribed.

1.6 When to Contact a Medical Professional

Call 911 or the local emergency number immediately if the person:

- Does not respond to you
- Is not breathing
- Has sudden chest pain or other symptoms of a heart attack

1.7 Prevention

Adults should take steps to control heart disease risk factors whenever possible.

- If you smoke, quit. Smoking more than doubles the chance of developing heart disease.
- Keep blood pressure, cholesterol, and diabetes in good control and follow your health care provider's orders.
- Lose weight if obese or overweight.
- Get regular exercise to improve heart health. (Talk to your provider before starting any new fitness program.)
- Eat a heart-healthy diet. Limit saturated fats, red meat, and sugars. Increase your intake of chicken, fish, fresh fruits and vegetables, and whole grains. Your provider can help you tailor a diet specific to your needs.

- Limit the amount of alcohol you drink. One drink a day is associated with reducing the rate of heart attacks, but two or more drinks a day can damage the heart and cause other medical problems.

2. First aid for burns

- ❖ **Stop the burning process** as soon as possible. This may mean removing the person from the area, dousing flames with water, or smothering flames with a blanket. Do not put yourself at risk of getting burnt as well.
- ❖ **Remove any clothing or jewelry** near the burnt area of skin, including babies' nappies. But do not try to remove anything that's stuck to the burnt skin, as this could cause more damage.
- ❖ **Cool the burn with cool or lukewarm running water** for 20 minutes as soon as possible after the injury. Never use ice, iced water, or any creams or greasy substances like butter.
- ❖ **Keep yourself or the person warm.** Use a blanket or layers of clothing, but avoid putting them on the injured area. Keeping warm will prevent **hypothermia**, where a person's body temperature drops below 35C (95F). This is a risk if you're cooling a large burnt area, particularly in young children and elderly people.
- ❖ **Cover the burn with cling film.** Lay the cling film over the burn, rather than wrapping it around a limb. A clean, clear plastic bag can be used for burns on your hand.
- ❖ **Treat the pain from a burn with: Paracetamol or ibuprofen.** Always check the manufacturer's instructions when using over-the-counter medication. Children under 16 years of age should not be given aspirin.
- ❖ **Raise the affected area, if possible.** This will help to reduce swelling.



2.1 When to go to hospital

Once you have taken these steps, you'll need to decide whether further medical treatment is necessary.

Go to a hospital **accident and emergency (A&E)** department for:

- large burns bigger than the size of the affected person's hand
- deep burns of any size that cause white or charred skin
- burns on the face, neck, hands, feet, any joints, or genitals
- all chemical and electrical burns
- any other injuries that need treating
- any signs of shock – symptoms include cold, clammy skin, sweating, rapid, shallow breathing, and weakness or dizziness

You should also go to hospital if you or the person who has been burned:

- is under the age of 10

- has a medical condition, such as heart, lung or liver disease, or **diabetes**
- has a weakened immune system (the body's defence system) – for example, because of **HIV or AIDS**, or because they're having **chemotherapy** for cancer

If someone has breathed in smoke or fumes, they should also get medical attention at hospital.

Some symptoms may be delayed, and can include:

- coughing
- a sore throat
- difficulty breathing
- singed nasal hair
- facial burns

You should also go to A&E if you have been injured intentionally, or think someone has been injured intentionally.

2.2 Electrical burns

Electrical burns may not look serious, but they can be very damaging. Someone who has an electrical burn should seek immediate medical attention at an A&E department.

If the person has been injured by a low-voltage source (up to 220 to 240 volts), such as a domestic electricity supply, safely switch off the power supply or remove the person from the electrical source using a material that doesn't conduct electricity, such as a wooden stick or a wooden chair.

Do not approach a person who's connected to a high-voltage source (1,000 volts or more).

2.3 Acid and chemical burns

Acid and chemical burns can be very damaging and require immediate medical attention at an A&E department. If possible, find out what chemical caused the burn and tell the healthcare professionals at A&E.

If you're helping someone else, put on appropriate protective clothing, such as gloves, and then:

- remove any contaminated clothing on the person
- if the chemical is dry, brush it off their skin

- use running water to remove any traces of the chemical from the burnt area

2.4 Sunburn:

You should relocate into the shade or, ideally, indoors if you observe any signs of sunburn, such as hot, red, or painful skin.

- For 10 minutes, take a cool bath or shower to soothe the burned skin.
- To hydrate, chill, and soothe the affected region, apply after-sun lotion. Use non-greasy or non-oily products.
- Ibuprofen or Paracetamol should help ease any pain you may be experiencing. Never administer aspirin to anybody under the age of 16, and always read the manufacturer's instructions.
- Keep an eye out for symptoms of heat exhaustion or heatstroke, which occur when your body temperature reaches 37 to 40C (98.6 to 104F) or higher. Dizziness, a fast heartbeat, or vomiting are symptoms.
- A person suffering from heat exhaustion should begin to feel better in about 30 minutes if they are swiftly transported to a cool location, given water to drink, and have their clothing loosen.

If they don't, heat stroke might occur. You must dial 999 in order to request an ambulance because this is a medical emergency.

3. First aid for Injury

When someone is injured, knowledge is the first line of defense so it's important to know how to treat basic injuries to avoid long-term and/or permanent damage. To help prepare you, we compiled a list of basic first-aid procedures for common injuries.

These treatments will use basic first aid techniques and supplies. If you're not sure what you should have in your first-aid kit, we compiled a list of items everyone should have, plus additional items for people that care for small children and those that use their kit outdoors.

3.1 Basic First Aid Treatment

Remember that when administering first aid treatment that involves contact with bodily fluids, it's important to wear protective gloves to eliminate the risk of disease transmission. For treatment of other injuries, gloves are optional.

3.2 Minor Wounds

Clean the injured area with soap and water and blot the wound dry. Avoid cleaning wounds with hydrogen peroxide or isopropyl alcohol as it can actually damage the tissue and delay healing. Once the wound is dry, apply antibiotic ointment and cover with gauze or a bandage.

3.3 Minor Cuts & Scrapes

First, stop the bleeding. If the cut or scrape does not stop bleeding on its own, apply pressure to the wound with clean a clean cloth or clean gauze while keeping the wound elevated. Clean the injury using soap and warm water, apply antibiotic ointment and a bandage, and watch for signs of infection.

3.4 Bee & Wasp Stings

First, try to remove the stinger with tweezers or by scraping the stung area with a credit card. Then apply a cool compress to reduce swelling and remove tight clothing or accessories from the area in case swelling occurs. Watch for signs of shock or allergic reaction by checking the victim's airway, breathing, and circulation. If any of these are impeded, call 911 immediately.

3.5 Nosebleeds

The first thing to remember with nosebleeds is that even though they are scary, they are rarely serious. It's important to stay calm, especially if the person with a nosebleed is a child, so as not to cause panic.

To stop a nosebleed, keep the person upright with the head tilted slightly forward. Have the person pinch the lower half of their nose (the soft part) and hold it firmly for ten minutes. After ten minutes, if bleeding persists, repeat the exercise.

3.6 Wrist & Ankle Sprains

Sprains happen when the ligaments in your ankle or wrist get stretched or torn. Most of the time these mild and can be treated at home. For wrist sprains, remember the acronym RICE: rest, ice, compression, elevation. Rest your wrist by using it as little as possible, apply ice in 20 minute

increments, wear an elastic compression wrap/bandage to help reduce swelling, and keep your wrist elevated at or above the level of your heart.

The at-home treatment for ankle sprains is similar. Remember the acronym MICE: motion, ice, compression, elevation. Motion may seem counterintuitive for a sprained ankle but it's important to try slowly flexing your foot upwards immediately after the sprain. Repeating this motion, combined with ice, compression, and elevation, can help reduce swelling and promote healing.

3.7 Fractures:

Fractures are broken bones, and they can occur as a result of falls or other harsh impacts. When this happens, the affected part should be immobilized, and additional manipulation of the affected area should be avoided. Remember that a fracture could sever a blood vessel or a nerve if it is not immobilized, resulting in a much more severe injury. Immobilize the injured part, and transport the patient to the nearest hospital or medical clinic as soon as possible.

PRICE – Treatment for soft tissue injuries and closed fractures

P	Protect the injury (stop using the injured limb, pad to protect)	
R	Rest the injury	
I	Ice - apply a wrapped ice pack	
C	Comfortable support – apply a supportive bandage	
E	Elevate – to reduce swelling	

4. First Aid for Fainting

Fainting is a form of unconsciousness that is quick and brief, often due to low blood sugar or standing in one place for too long. Fainting can also be caused by a more serious medical matter.

Unconsciousness is an abnormal state in which a person is not alert and not fully responsive to his/her surroundings.

Unlike when a person is asleep, someone who has fainted and is unconscious cannot cough, clear his/her throat, or turn his/her head if in distress. When unconscious, a person is in danger of choking, making it very important to keep the airway clear while awaiting medical care.

4.1 First Aid Guide

In the event of fainting, attempt the following self-care measures:

- If possible, try to prevent someone who is fainting from hitting the ground.
 - Lay the person down on the ground, face up, and elevate his/her feet 8–12 inches.
 - Loosen any constrictive clothing.
 - Apply a cool, wet compress to the person's forehead.
 - Attempt to keep the person from standing up until fully recovered.
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- **Note:**
 - If the person vomits while he/she is unconscious, quickly turn him/her to allow the fluid to drain while protecting the person's airway.
 - Do not attempt to give an unconscious person anything by mouth.
 - Do not shake or slap a person who has fainted to attempt to make him/her regain consciousness.

If the person was injured while fainting, give first aid for any injuries (eg, bumps, bruises, or cuts) accordingly. Bleeding should be stopped with direct pressure.



4.2 Who's at risk?

People with the following medical conditions are at an increased risk of fainting:

- Low blood sugar (hypoglycemia)
- Anemia
- Any condition in which there is a rapid loss of blood, such as from internal bleeding
- Heart and circulatory problems
- Heat stroke or heat exhaustion
- Eating disorders (e.g., anorexia and bulimia)
- Toxic shock syndrome

4.3 Signs and Symptoms

A person who faints often experiences dizziness, nausea, weakness, and/or blurred vision prior to losing consciousness, and they may have cold, clammy, pale skin. The person may be disoriented, incoherent, motionless, and silent.

4.4 When to Seek Medical Care

If the person does not regain consciousness quickly (e.g., within 5 minutes), is elderly, or has a medical problem, call emergency medical services. While not typical, fainting can be a sign of serious illness, so seek treatment by a medical professional to determine the cause and provide proper treatment.

Conclusions

At some point in everyone's life, he/she will receive or give first aid. Early response can increase the likelihood of a favorable outcome of an emergency. It is important to help the victim as much as possible, but early notification to the proper medical professionals in your community will ensure the best outcome for any victim. Remember to always ensure scene safety and dial 911 when necessary before assisting a victim.

References

<https://www.realsimple.com/health/first-aid-health-basics/basic-first-aid>

<https://www.safeopedia.com/2/1124/injuries-and-diseases/injuries/first-aid-for-the-10-most-common-workplace-injuries>

<https://www.webmd.com/first-aid/fainting-treatment>